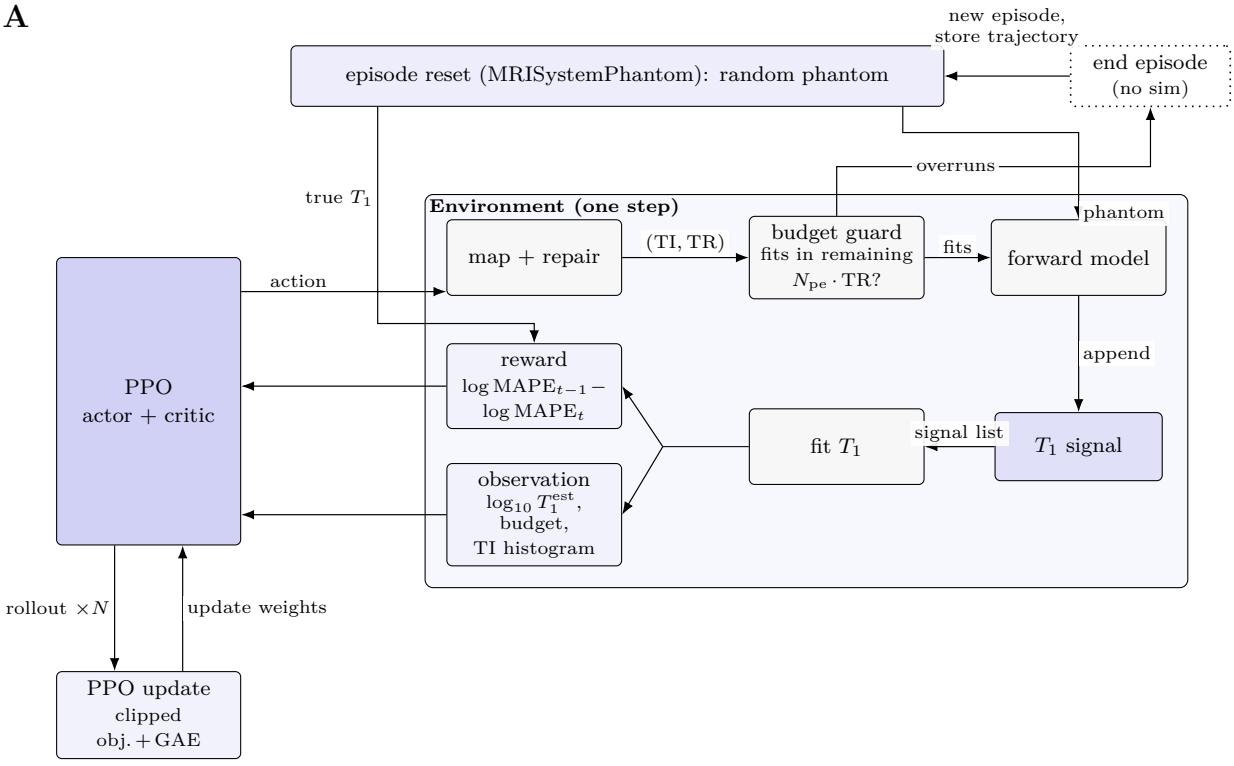
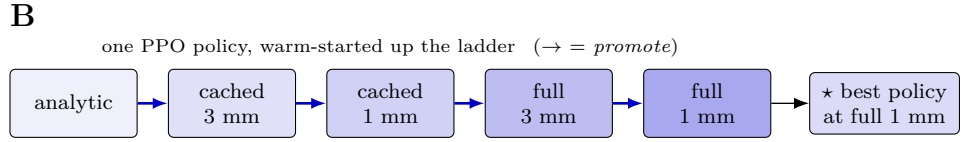
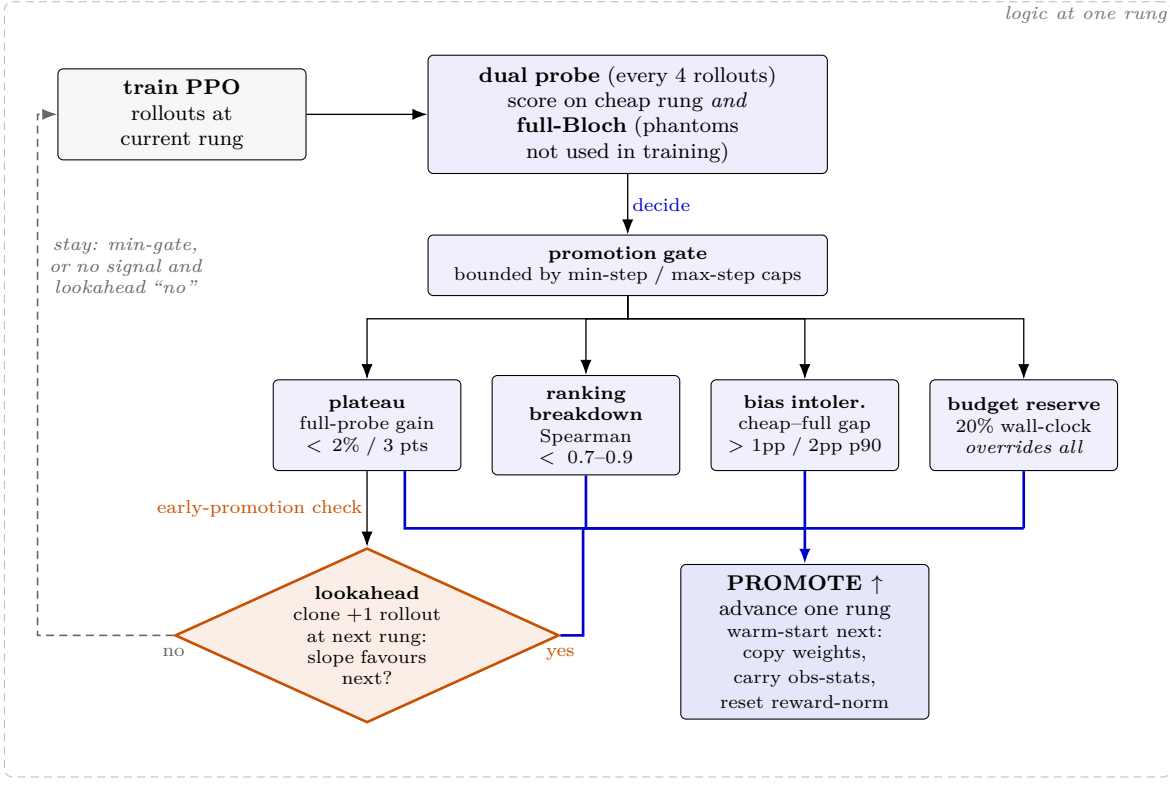

A
episode reset (MRISystemPhantom): random phantom
new episode, store trajectory
end episode (no sim)
overruns
true T_1
Environment (one step)
map + repair
(TI, TR)
budget guard fits in remaining $N_{pe} \cdot TR$?
fits
forward model
phantom
action
PPO actor + critic
reward $\log \text{MAPE}_{t-1} - \log \text{MAPE}_t$
append
fit T_1
signal list
T_1 signal
observation $\log_{10} T_1^{est}$, budget, TI histogram
rollout $\times N$
update weights
PPO update clipped obj. + GAE


B
one PPO policy, warm-started up the ladder (→ = promote)
analytic
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm
⋆ best policy at full 1 mm


logic at one rung
train PPO rollouts at current rung
dual probe (every 4 rollouts) score on cheap rung and full-Bloch (phantoms not used in training)
decide
promotion gate bounded by min-step / max-step caps
stay: min-gate, or no signal and lookahead "no"
plateau full-probe gain < 2% / 3 pts
ranking breakdown Spearman < 0.7–0.9
bias intoler. cheap–full gap > 1pp / 2pp p90
budget reserve 20% wall-clock overrides all
early-promotion check
lookahead clone +1 rollout at next rung: slope favours next?
no
yes
PROMOTE ↑ advance one rung warm-start next: copy weights, carry obs-stats, reset reward-norm